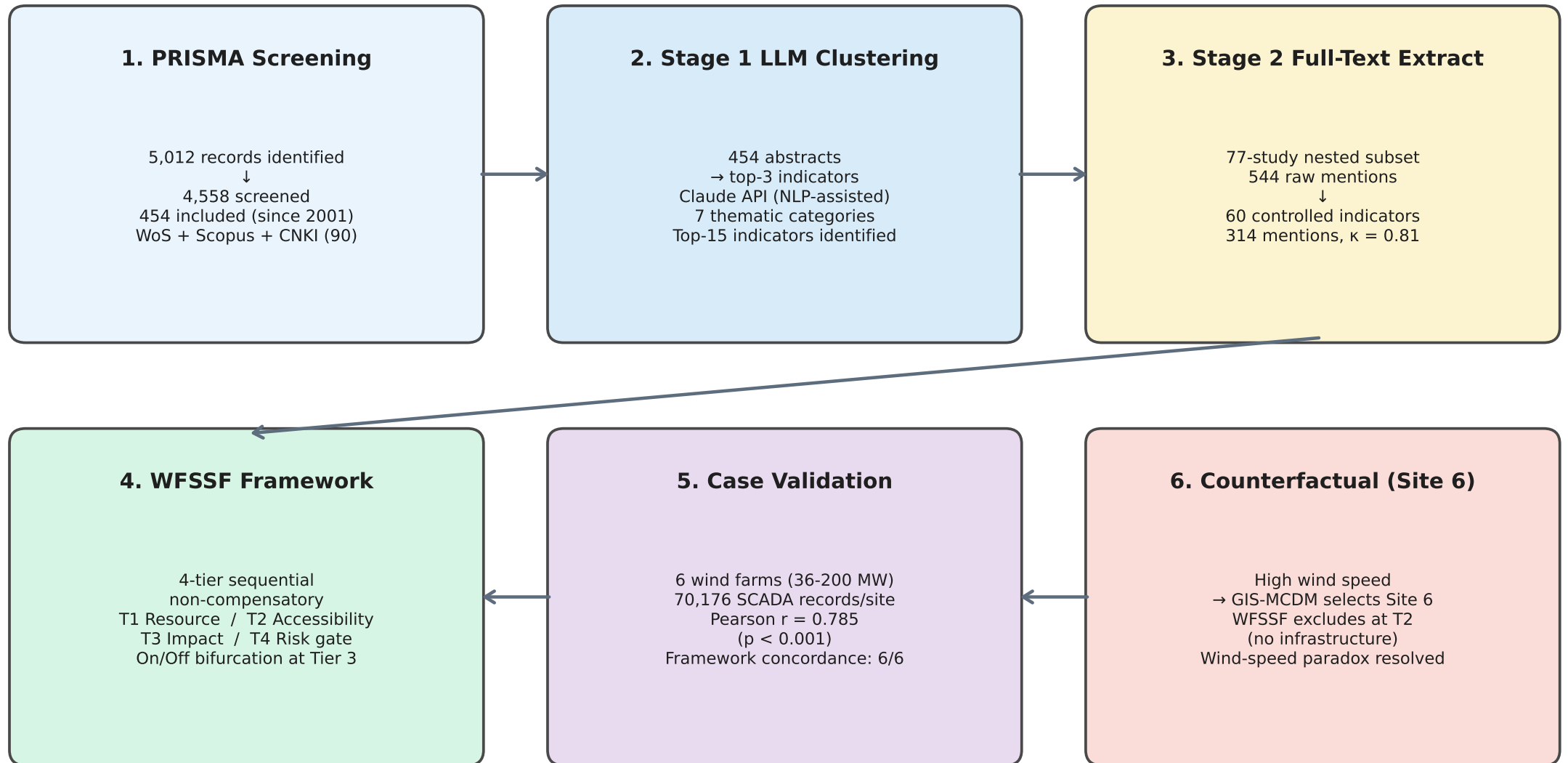




Data flow : 5,012 records → 454 abstracts → 77 full-text → 60 indicators → 4-tier framework → 6-site validation

Output : 60-term controlled vocabulary, 4-tier WFSSF, 6-site empirical concordance, counterfactual failure-mode catalogue